

AI Human Activity Recognition (HAR)

for On-board BAS Experiments — explained in plain English

This document walks through the problem statement end to end — what it means, why it matters, and how we're planning to approach it — in plain language, so everyone on the team starts from the same page before we split up the work.

1. The Story — What Problem Are We Actually Solving?

BAS stands for Bharatiya Antariksh Station — India's planned space station. Along with future Moon missions, astronauts on board will be running real scientific experiments: biology, materials science, fluid physics, and so on.

Right now, if an astronaut on the ISS isn't sure they did a step correctly, mission control on Earth can watch a video feed and talk them through it almost instantly, because the ISS is close to Earth.

The catch:

Once you go further out — the Moon, or beyond — there's a real communication delay. Messages take seconds to minutes to travel each way. If an astronaut is mid-experiment and needs a "yes, that's correct, go to the next step" from Earth, they might not get it in time. Some experiments are time-sensitive — a reaction might need the next step within 90 seconds. Waiting for Earth just isn't an option anymore.

So the ask is: build an AI system that lives **on board** the spacecraft, watches the astronaut while they work, recognizes what physical action they're doing, and checks it against the correct step-by-step sequence of the experiment — without needing to phone home to Earth.

Think of it like a very attentive lab assistant standing next to the astronaut, silently checking off each step as it happens, and gently flagging it if a step is done wrong, skipped, or done out of order.

2. Why This Isn't a Normal "Recognize the Action" App

Human Activity Recognition (HAR) already exists — it's the tech behind fitness trackers that know if you're walking, running, or sitting. But this problem is a much harder version of that, for three reasons:

a) It's not just "what action" — it's "is this the right action, right now"

A fitness tracker just labels an action. Here, the system needs to know the entire correct order of steps for the experiment, and catch it if the astronaut skips a step, repeats one, or does step 5 before step 3.

b) Space breaks the normal rules of movement

In microgravity, bodies float, objects drift instead of falling, and astronauts often anchor themselves with tethers or foot restraints. Movement looks and feels different from anything a model trained on ordinary Earth video has ever seen. Bulky suits and gloves also hide the small hand movements that most action-recognition models rely on.

c) It has to run on limited on-board hardware, reliably

There's no cloud GPU cluster in space. Whatever we build has to run on modest on-board compute, in real time, and it has to be trustworthy enough that astronauts actually rely on it during a real experiment — false alarms and missed errors are both costly.

3. How the System Would Actually Work (Big Picture)

It helps to think of the system as two layers stacked on top of each other:

1. Action Recognizer	Looks at the astronaut's movement over a short time window and labels what action is happening right now.	"They're currently opening the sample container."
2. Sequence Validator	Compares the stream of recognized actions against the experiment's correct step order.	"Wait — step 4 was supposed to happen before step 6. Flag it."

If the system isn't confident about what it's seeing, it shouldn't silently pass or fail the step — it should ask the astronaut for a quick confirmation instead. A wrong silent decision is more dangerous than a system that occasionally double-checks itself.

4. Initial Steps — Where We Actually Start

We shouldn't try to solve "recognize any experiment ever" on day one. Here's a realistic order of operations:

1

Pick one experiment and map it out step by step

Choose a single real or realistic experiment protocol and break it into a strict, ordered checklist of small physical actions — e.g. "unlock case → remove vial → attach to holder → press switch → wait 60 seconds → record reading → reseal." This checklist becomes our ground truth.

2

Choose what we're actually sensing

Decide between camera video, camera + pose/skeleton tracking (like MediaPipe), or wearable motion sensors (IMUs) on gloves/suit — or a mix. This choice affects everything downstream, so it needs to be an early decision, not an afterthought.

3

Solve the data problem first

There's no existing dataset of "astronauts performing BAS protocols." We'll need to record volunteers doing the exact protocol ourselves — from multiple angles, multiple people, and deliberately including wrong-order and wrong-action examples, not just correct runs.

4

Get closer to real microgravity motion if we can

Where possible, use analog environments — parabolic flight, neutral buoyancy pools, or astronaut training facilities — to capture motion that's closer to how people actually move in space. Simulated/synthetic data can help fill gaps cheaply too.

5

Build the two-layer model described above

Start with the action recognizer (a small temporal model over the pose/sensor data), then add the sequence validator (a simple state-machine style logic layer) on top.

6

Test for confidence, not just accuracy

Make sure the system knows when it's unsure and asks rather than guesses — this matters more in this use case than raw accuracy numbers.

7

Only then, optimize for the hardware

Once it works well offline, shrink the model (quantization, lightweight architectures) so it can actually run on modest on-board compute in real time.

5. TL;DR — The One-Paragraph Version

In one breath:

Astronauts on future deep-space missions can't rely on real-time help from Earth because of communication delay. So we're building an on-board AI that watches how an astronaut moves, figures out what step of a science experiment they're doing, and checks it against the correct order of steps — catching mistakes instantly, without needing to talk to Earth at all. First step: pick one experiment, map its steps, and start collecting motion data for it — the model comes after that.

6. Open Questions for the Team

- Which single experiment protocol should we use as our first demo case?
- Do we have access to any analog-gravity recording setup, or are we starting with normal-gravity volunteer recordings only?
- Camera-based, wearable-based, or a hybrid — what's realistic given our timeline and gear?
- What should happen when the system is uncertain — pause the astronaut, log a warning, or ask for manual confirmation?

Prepared as a plain-English walkthrough of the problem statement for internal team discussion.